

Start coding or [generate](#) with AI.

Variables

```
fruit = "mango"
print(fruit)
```

mango

Variable Naming

1. A variable name must start with a letter or an underscore.
2. A variable name can't start with a number.
3. A variable name can only contain alpha numeric characters (A-Z, 0-9, _)
4. Variable names are case sensitive.
5. A variable name can't be any of the python keyword.

```
fruit = "apple"
print(fruit)
```

apple

```
9fruit = "orange"
```

```
print(9fruit)
```

```
File "/tmp/ipython-input-2086616234.py", line 1
  9fruit = "orange"
    ^
SyntaxError: invalid decimal literal
```

```
Fruit_9 = "Kiwi"
print(Fruit_9)
```

Kiwi

```
Age = 30
```

```
age = 40
```

```
AGE = 50
```

```
print(Age, age, AGE)
```

30 40 50

```
def = "Apple"
```

```
print(def)
```

```
File "/tmp/ipython-input-2398966055.py", line 1
  def = "Apple"
    ^
SyntaxError: invalid syntax
```

```
a = 20
```

```
b = 30
```

```
print(a+b)
```

50

```
#Task1 : Perform Multiplication, division & Subtraction
```



```
fname = "Shubham"

lname = "Singh"

print(fname , '',lname)

Shubham Singh
```

Data Types

Text Type : str

Numeric Type : int, float , complex

Sequence Types : list , tuple , range

Mapping Type : dict

Set Types : set

Boolean Type : bool

```
a = "apple"
type(a)
```

str

```
b = 90
type(b)
```

int

```
c = 93.74
type(c)
```

float

```
d = 6+8j
type(d)
```

complex

List

1. List is used to store multiple items in it.
2. It is mutable.
3. It is ordered.
4. It allows duplicate values.
5. It is denoted by []

```
fruit = ["apple", "mango", "banana", 2, 34, 8.34, True, "apple", "banana"]
print(fruit)
type(fruit)
```

```
['apple', 'mango', 'banana', 2, 34, 8.34, True, 'apple', 'banana']
list
```

```
fruit[0]
```

'apple'

```
fruit[-2]
```

'apple'

```
fruit[0:5]
```

['apple', 'mango', 'banana', 2, 34]

```
#Task2 : Perform Negative Indexing
```

```
fruit.append("kiwi")
print(fruit)
```

```
['apple', 'mango', 'banana', 2, 34, 8.34, True, 'apple', 'banana', 'kiwi']
```

```
fruit.insert(2,"orange")
print(fruit)
```

```
['apple', 'mango', 'orange', 'banana', 2, 34, 8.34, True, 'apple', 'banana', 'kiwi']
```

```
fruit[2]="Papaya"
print(fruit)
```

```
['apple', 'mango', 'Papaya', 'banana', 2, 34, 8.34, True, 'apple', 'banana', 'kiwi']
```

```
fruit.remove("Papaya")
print(fruit)
```

```
['apple', 'mango', 'banana', 2, 34, 8.34, True, 'apple', 'banana', 'kiwi']
```

```
fruit.pop(2)
print(fruit)
```

```
['apple', 'mango', 2, 34, 8.34, True, 'apple', 'banana', 'kiwi']
```

```
fruit.clear()
print(fruit)
```

```
[]
```

```
del fruit
print(fruit)
```

```
-----
NameError                                Traceback (most recent call last)
/tmp/ipython-input-115828921.py in <cell line: 0>()
      1 del fruit
----> 2 print(fruit)

NameError: name 'fruit' is not defined
```



Tuple

1. Tuple is used to store multiple items in it
2. It is immutable.
3. It is ordered.
4. It allows duplicate values.
5. It is defined by ().

```
fruit = ('apple','mango','orange',12,94.43,True,'apple','mango')
print(fruit)
type()
```

```
('apple', 'mango', 'orange', 12, 94.43, True, 'apple', 'mango')
tuple
```

```
fruit[5] #Indexing
```

```
True
```

```
fruit[0:4]
```

```
('apple', 'mango', 'orange', 12)
```

```
fruit[0:7:2]
```

```
('apple', 'Kiwi', 94.43, 'apple')
```

```
fruit[-5:-1]
```

```
(12, 94.43, True, 'apple')
```

```
fruit = ('apple','mango','orange',12,94.43,True,'apple','mango')
x = list(fruit) #Type Casting
print(x)
x[2] = "Kiwi"
fruit = tuple(x)
print(fruit)
```

```
['apple', 'mango', 'orange', 12, 94.43, True, 'apple', 'mango']
('apple', 'mango', 'Kiwi', 12, 94.43, True, 'apple', 'mango')
```

```
#Task3 : Perform append,insert,pop,remove,clear & delete as well
```

Set

1. Set is also used to store multiple items in it.
2. It doesn't allow duplicate values.
3. It is unordered.
4. It is immutable.
5. It is defined by {}



```
fruit = {'apple', 'mango', 'orange', 12, 94.43, True, 'apple', 'mango'}
print(fruit)
type(fruit)
```

```
{True, 'orange', 'mango', 12, 94.43, 'apple'}
set
```

```
for i in fruit:
    print(i)
```

```
True
orange
mango
12
94.43
apple
```

```
fruit.add("Papaya")
print(fruit)
```

```
{True, 'orange', 'Papaya', 'mango', 12, 94.43, 'apple'}
```

```
fruit.remove("apple")
print(fruit)
```

```
{True, 'orange', 'Papaya', 'mango', 12, 94.43}
```

```
fruit.discard("mango")
print(fruit)
```

```
{True, 'orange', 'Papaya', 12, 94.43}
```

```
fruit.remove("apple")
print(fruit)
```

```
-----
KeyError                                Traceback (most recent call last)
/tmp/ipython-input-3525016582.py in <cell line: 0>()
----> 1 fruit.remove("apple")
      2 print(fruit)

KeyError: 'apple'
```

```
fruit.discard("mango")
print(fruit)
```

```
{True, 'orange', 'Papaya', 12, 94.43}
```

```
fruit.clear()
print(fruit)
```

```
set()
```

```
del fruit
print(fruit)
```

```
-----
NameError                                Traceback (most recent call last)
/tmp/ipython-input-1215277667.py in <cell line: 0>()
      1 del fruit
----> 2 print(fruit)

NameError: name 'fruit' is not defined
```



Dictionary

1. Dictionary is used to store items in a key value pair.
2. It is ordered*.
3. It can have duplicate values but not the keys
4. It is mutable*.
5. It is denoted by {}

```
car = {"brand": "Ford",
       "model": "Mustang",
       "year": 1964}
print(car)
type(car)
```

```
{'brand': 'Ford', 'model': 'Mustang', 'year': 1964}
dict
```

```
print(car['model'])
```

```
Mustang
```

```
car.keys()
```

```
dict_keys(['brand', 'model', 'year'])
```

```
car.values()
```

```
dict_values(['Ford', 'Mustang', 1964])
```

```
#Updating the value
car['brand'] = 'Maruti'
print(car)
```

```
{'brand': 'Maruti', 'model': 'Mustang', 'year': 1964}
```

```
car.update({'year':2022})  
print(car)
```

```
{'brand': 'Maruti', 'model': 'Mustang', 'year': 2022}
```

#Adding new key:value pair

```
car['color']='red'  
print(car)
```

```
{'brand': 'Maruti', 'model': 'Mustang', 'year': 2022, 'color': 'red'}
```

```
car.update({'engine_type':'petrol'})  
print(car)
```

```
{'brand': 'Maruti', 'model': 'Mustang', 'year': 2022, 'color': 'red', 'engine_type': 'petrol'}
```

```
car.popitem() #removes last inserted item  
print(car)
```

```
{'brand': 'Maruti', 'model': 'Mustang', 'year': 2022, 'color': 'red'}
```

```
car.pop('year')  
print(car)
```

```
{'brand': 'Maruti', 'model': 'Mustang', 'color': 'red'}
```

```
del car  
print(car)
```

```
-----  
NameError                                Traceback (most recent call last)  
/tmp/ipython-input-3530790737.py in <cell line: 0>()  
      1 del car  
----> 2 print(car)  
  
NameError: name 'car' is not defined
```

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